PO11Q - Quantitative Political Analysis:
From Measurement to Inference

Week 9, Exercise Solutions

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The t-distribution

1. How df changes the shape and critical values
   Using two-tailed, $\alpha = 0.05$ (so $t_{crit} = t_{0.025,df}$ with $df = n - 1$):
   - $n = 3$ ($df = 2$): $t_{crit} \approx 4.303$
   - $n = 5$ ($df = 4$): $t_{crit} \approx 2.776$
   - $n = 10$ ($df = 9$): $t_{crit} \approx 2.262$
   - $n = 20$ ($df = 19$): $t_{crit} \approx 2.093$
   - $n = 50$ ($df = 49$): $t_{crit} \approx 2.009$

   The normal reference is $z_{0.025} \approx 1.960$.
   Within 0.05 of 1.96 occurs by $n \approx 50$ ($df = 49$, 2.009 is 0.049 above 1.96).

   Explanation: as $df \to \infty$, the t distribution $\to N(0, 1)$, so tails thin and t_{crit} decreases toward 1.96.

2. One-tailed vs two-tailed critical regions
   With $df = 14$ and $\alpha = 0.05$:
   - Two-tailed: $t_{0.025,14} \approx 2.145$, cut points at ± 2.145
   - One-tailed (upper): $t_{0.05,14} \approx 1.761$, single cut point at $+1.761$

   For $H_0 : \mu = \mu_0$ vs $H_1 : \mu > \mu_0$, the one-tailed threshold $t_{0.05,14}$ is relevant because only the upper tail provides evidence against H_0.

   In both cases, the shaded area equals α:
   - two-tailed shades two symmetric regions each of area $\alpha/2$;
   - one-tailed shades a single upper tail of area α.